

```
$ helm install --dry-run --debug todo-app-deploy ./todo
install.go:159: [debug] Original chart version: ""
install.go:176: [debug] CHART PATH: /Volumes/data/src/playbooks/k8s/helm/todo

NAME: todo-app-deploy
LAST DEPLOYED: Fri Sep  4 18:30:15 2020
NAMESPACE: default
STATUS: pending-install
REVISION: 1
USER-SUPPLIED VALUES:
{}

COMPUTED VALUES:
affinity: {}
autoscaling:
  enabled: false
  maxReplicas: 100
  minReplicas: 1
  targetCPUUtilizationPercentage: 80
fullnameOverride: ""
image:
  pullPolicy: IfNotPresent
  repository: quay.io/akasurde/todo
  tag: latest
```

Figure 3: Deployment dry run

```
$ helm install todo-app-deploy ./todo
NAME: todo-app-deploy
LAST DEPLOYED: Fri Sep  4 18:32:45 2020
NAMESPACE: default
STATUS: deployed
REVISION: 1
NOTES:
1. Get the application URL by running these commands:
  export POD_NAME=$(kubectl get pods --namespace default -l "app.kubernetes.io/name=todo,app.kubernetes.io/instance=todo-app-deploy" -o jsonpath="{.items[0].metadata.name}")
  echo "Visit http://127.0.0.1:8080 to use your application"
  kubectl --namespace default port-forward $POD_NAME 8080:80
```

Figure 4: Final deployment

```
$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-minikube-797f975945-kj5kg     1/1     Running   5           94d
todo-app-deploy-659bfbd7f-x7sb9     1/1     Running   0           35s

$ export POD_NAME=$(kubectl get pods --namespace default -l "app.kubernetes.io/name=todo,app.kubernetes.io/instance=todo-app-deploy" -o jsonpath="{.items[0].metadata.name}")
$ echo "Visit http://127.0.0.1:8080 to use your application"
Visit http://127.0.0.1:8080 to use your application
$ kubectl --namespace default port-forward $POD_NAME 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
```

Figure 5: Output

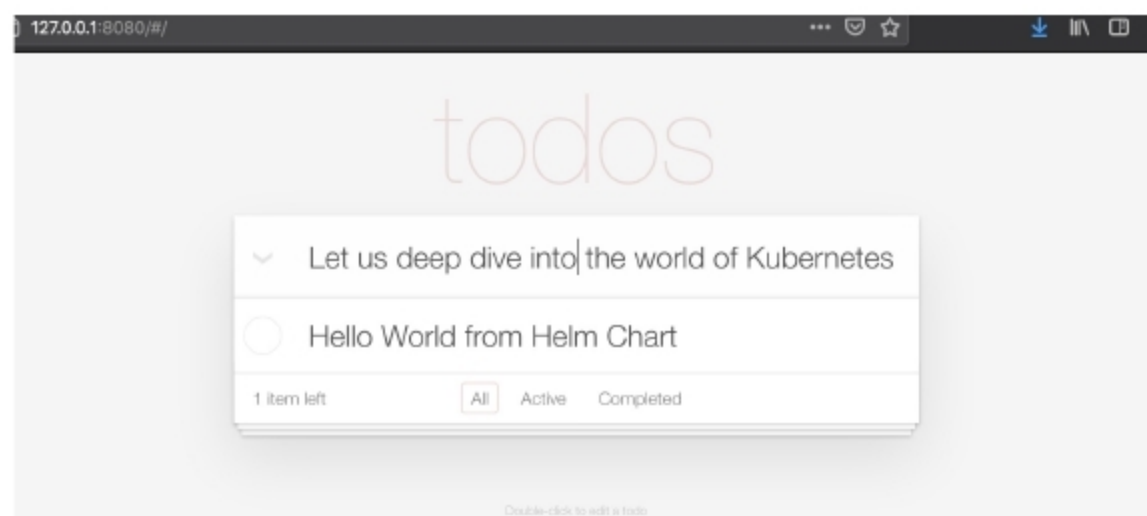


Figure 6: Application in action

Before proceeding to actual deployment, we can use *--dry-run* to debug and inspect the actual deployment (see Figure 3).

After inspection and verifying the deployment, you can deploy Chart, as shown in Figure 4.

Follow the instructions provided at the end of the command (see Figure 5).

If you visit <http://127.0.0.1:8080>, you can see the *todo* application.

Let us understand the basics behind this. This application is a *todo* application written in React from <http://todomvc.com/>, containerised and hosted at <https://quay.io/repository/akasurde/todo>.

This container image value is specified in your *values.yml* as a repository, and tagged in the image section. The pod is also exposed to port 8080 using the *todo/templates/service.yaml* file under the templates directory.

Packaging Chart

So far, we have been dealing with the unpackaged Chart. In order to distribute it to the users or the team, you can use the *package* sub-command.

```
$ helm package ./todo
```

Successfully packaged chart and saved it to:
</data/helm/todo-0.1.0.tgz>

You can now distribute this tarball to your team or users. Users can install this tarball as:

```
$ helm install todo-app-latest todo-0.1.0.tgz
```

In the next article on Helm, we will see how you can setup your own private Helm Chart repository to distribute Helm Chart. **END** 🐧

References

- [1] Helm Github page: <https://github.com/helm/helm/>
- [2] TodoMVC framework: <http://todomvc.com/>
- [3] Quay.io: <https://quay.io/>

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